Angular Advanced 02 – Lazy loading

Peter Kassenaar –
info@kassenaar.com

What is lazy loading

- Deferred loading of modules, until the user needs them.
  - OR: for optimal user experience:
  - Load the minimum setup for the application to work, so the user can interact with the app.
  - Then asynchrounously load other modules.
  - They are instantly available if the user navigates to them
- Only *modules* can be loaded lazily, not *components*.
- Lazy loading works in conjunction with the router.
- It is considered best practice nowadays to use LL from the start

Victor Savkin – creator of the router

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Victor Savkin

Co-founder of Narwhal Technologies (nrwl.io), where we provide Angular consulting to large teams who wa...

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Angular Router: Preloading Modules

Victor Savkin is a co-founder of nrwl.io, providing Angular consulting to enterprise teams. He was previously on the Angular core team at Google, and built the dependency injection, change detection, forms, and router modules.

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https://vsavkin.com/angular-router-preloading-modules-ba3c75e424cb

Official documentation

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`queryParamsHandling` and `preserveFragment` bindings respectively.

Milestone 6: Asynchronous routing

As you've worked through the milestones, the application has naturally gotten larger. As you continue to build out feature areas, the overall application size will continue to grow. At some point you'll reach a tipping point where the application takes long time to load.

How do you combat this problem? With asynchronous routing, which loads feature modules *lazily*, on request. Lazy loading has multiple benefits.

- You can load feature areas only when requested by the user.
- You can speed up load time for users that only visit certain areas of the application.
- You can continue expanding lazy loaded feature areas without increasing the size of the initial load bundle.

You're already made part way there. By organizing the application into modules— `AppModule` , `HeroesModule` , `AdminModule` and `CrisisCenterModule` —you have natural candidates for lazy loading.

Some modules, like `AppModule` , must be loaded from the start. But others can and should be lazy loaded. The `AdminModule` , for example, is needed by a few authorized users, so you should only load it when requested by the right people.

Lazy Loading route configuration

Change the `admin` path in the `admin-routing.module.ts` from `'admin'` to an empty string, `''` , the *empty path*.

The `Router` supports *empty path* routes; use them to group routes together without adding any additional path segments to the URL. Users will still visit `/admin` and the `AdminComponent` still serves as the *Routing*

authentication
Component-less route: grouping routes without a component
CanActivateChild: guarding child routes
CanDeactivate: handling unsaved changes
Cancel and save
Resolve: pre-fetching component data
Fetch data before navigating
Query parameters and fragments
Milestone 6: Asynchronous routing
Lazy Loading route configuration
CanLoad Guard: guarding unauthorized loading of feature modules
Preloading: background loading of feature areas
Custom Preloading Strategy
Inspect the router's configuration
Wrap up and final app
Appendices
Appendix: link parameters array

https://angular.io/guide/router#asynchronous-routing

How to lazy load

Add or edit `app.routing.module.ts`

- Don't point directly to components
- Point to Modules instead. Use `loadChildren()`

```
const routes: Routes = [
   {path: '', redirectTo: 'customers', pathMatch: 'full'},
   {path: 'customers', loadChildren: './customer/customer.module#CustomerModule'},
   {path: 'products', loadChildren: './products/products.module#ProductsModule'},
];

export const AppRoutingModule = RouterModule.forRoot(routes);
```

Edit `app.module.ts` (no more loading of modules)

```
// import routing module that defines the LL
import {AppRoutingModule} from './app.routing.module';

@NgModule({
    …
   imports      : [
      BrowserModule,
      AppRoutingModule
   ],
   bootstrap    : [AppComponent]
})
export class AppModule {
}
```

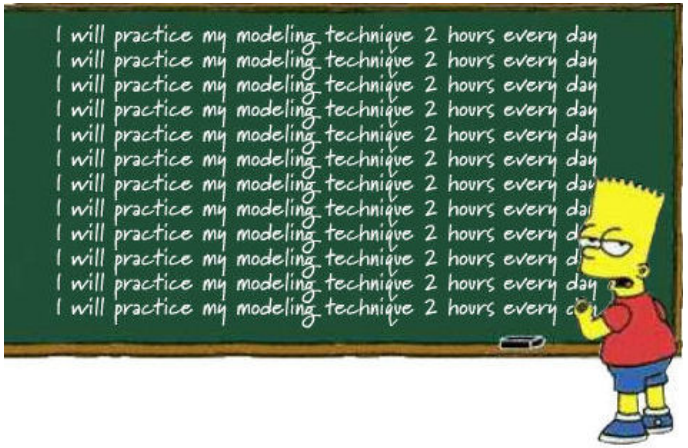
Edit separate modules,
add `RouterModule.forChild()` with various components.

```
import {RouterModule, Routes} from '@angular/router';

// Lazy loaded routes for this module
const customerRoutes: Routes = [
   {path: '', component: CustomerComponent}
];

@NgModule({
   imports     : [
      …
      RouterModule.forChild(customerRoutes)
   ],
    …
})
export class CustomerModule {
}
console.log('CustomerModule loaded lazily...');
```

Workshop

- Open `../110-lazy-loading`.
- Create a new module
- Create a new component inside this new module and give it some UI.
- Add a route to the new component
- Use the new module in the root module and lazy load it
- Add a link to navigate to the lazy loaded module.

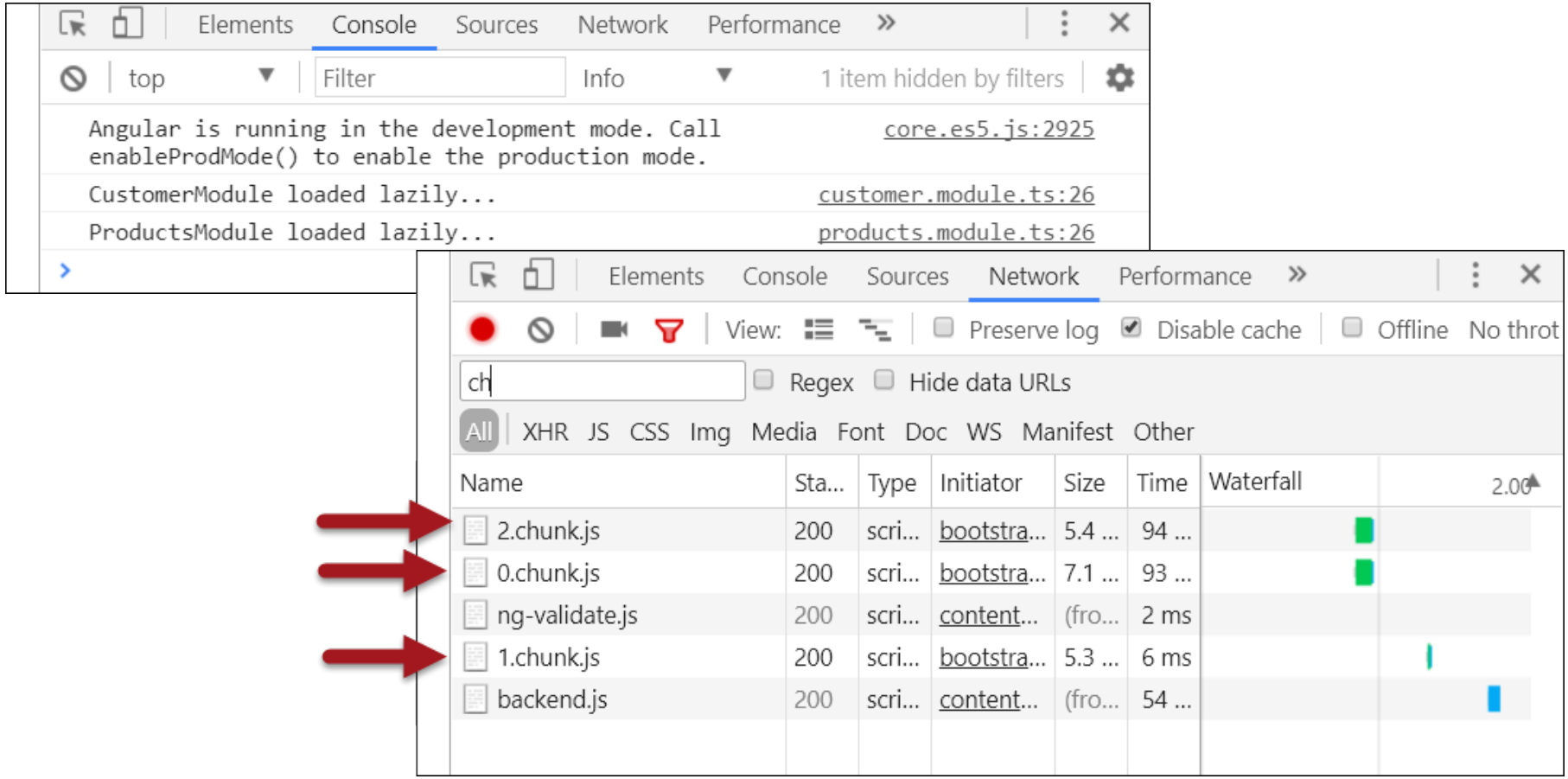
- *OR:*
- Add LL from scratch to your own application, using the steps described in this module.



Preloading Strategies

- Optimize Lazy Loading even further: preloading strategies
  - Load all modules in background
  - Load only modules *you want to load* in the background
- Default preloading: PreloadAllModules

```
import {ExtraOptions, PreloadAllModules,
        RouterModule, Routes} from '@angular/router';


const config: ExtraOptions    = {
   preloadingStrategy: PreloadAllModules
};

export const AppRoutingModule = RouterModule.forRoot(routes, config);
```



https://angular.io/api/router/PreloadAllModules

Custom preloading strategy

- Define which module(s) are loaded lazily, while others are loaded on demand

- Solution: compose a strategy that *only* preloads routes when a custom `data.preload` flag is set to `true`

```
{
    path        : 'products',
    loadChildren: './products/products.module#ProductsModule',
    data        : {preload: true}
},
{
    path        : 'big-module',
    loadChildren: './very-big-module/very-big-module.module#VeryBigModule'
},
```

Steps

1. Create new module, with a (potential) heavy load
2. Add `data` property and set `{ preload:true }` to every route you want to load lazily
3. Assign custom preloader to `preloadingStrategy`:

```
…
const config: ExtraOptions = {
   preloadingStrategy: MyCustomPreloader
};
@NgModule({
   imports   : [RouterModule.forRoot(routes, config)],
   exports   : [RouterModule],
   providers: [MyCustomPreloader]
})
export class AppRoutingModule {
}
```

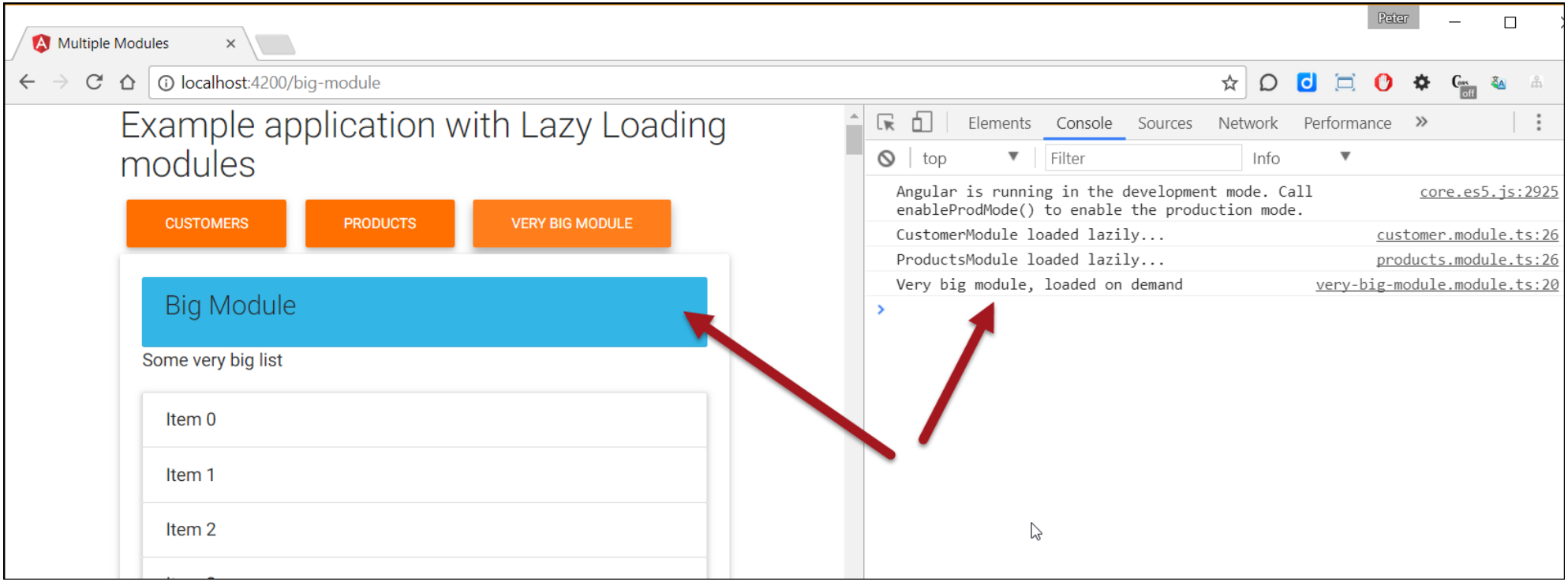
Define custom loader

```
// app.routing.loader.ts
import { PreloadingStrategy, Route } from '@angular/router';

import { Observable } from 'rxjs/Observable';
import 'rxjs/add/observable/of';

export class MyCustomPreloader implements PreloadingStrategy {
   preload(route: Route, load: Function): Observable<any> {
      // only preload the route if data attribute is set and preload===true
      return route.data && route.data.preload ? load() : Observable.of(null);
   }
}
```

Official documentation https://angular.io/guide/router#custom-preloading-strategy

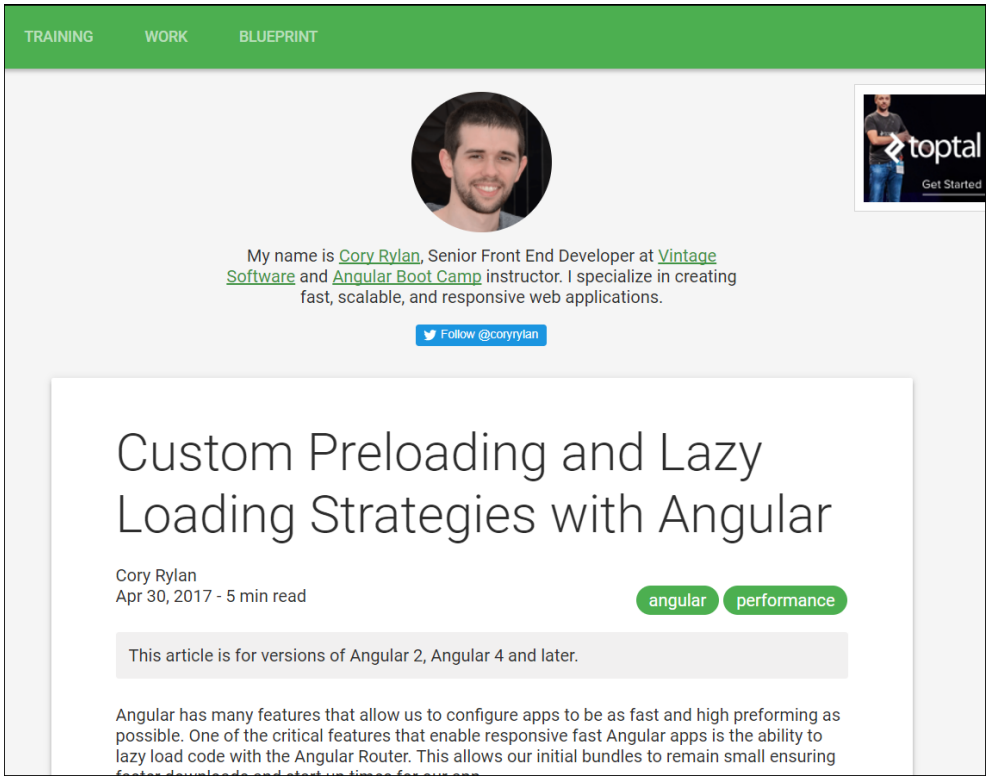
Run the app

Run the app. The first 2 modules should be loaded lazily, the third module should be loaded on demand



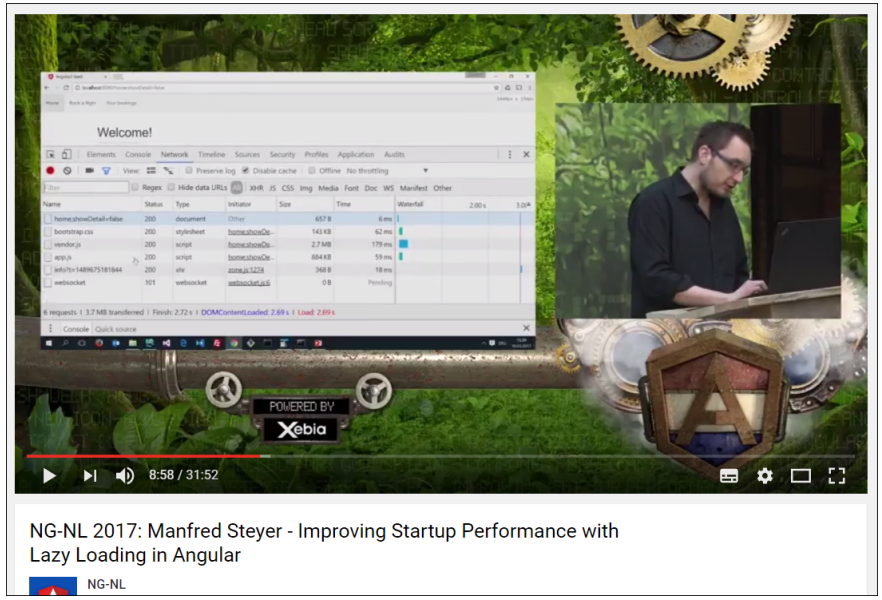
Example: ../120-custom-preloading

More information

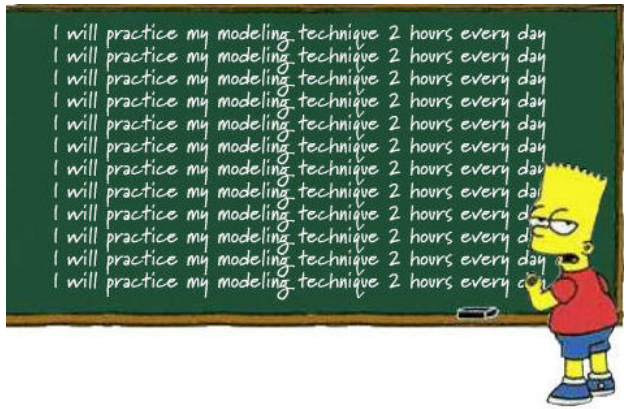


https://coryrylan.com/blog/custom-preloading-and-lazy-loading-strategies-with-angular

Manfred Steyer - Improving Startup Performance with Lazy Loading in Angular



https://www.youtube.com/watch?v=n6EMOeCDfjc

Workshop

- Add a new module w/ component to your application.
- Add the module to the routing section of your application. Add a link to navigate to the route.
- Let *other* components be loaded lazily by adding a `data` property
- Write a custom preloading class, in `app.preloader.ts`
- Add the custom preloader to `app.routing.module.ts`.
  - Note: make sure this is (now) actually a Module, as it has to import and provide `app.preloader.ts`